

Statistics refresher

Simulation in Prob and Stats (18284)

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S.1 Inference refresher

S.2 Statistical analysis of simulated data

S.3 Statistical validation techniques

S.1 Inference refresher

Probability models commonly depend on an unknown (scalar or vector-valued) *parameter*, θ , as F_θ . In order to approximate the value of the parameter, a random sample $X_1, \dots, X_n$ of $X \sim F_\theta$ is taken and an estimator of θ , $\hat{\theta}_n = \hat{\theta}(X_1, \dots, X_n)$ is evaluated over it.

- $\hat{\theta}_n$ is **unbiased** (as an estimator of θ) if $\mathbb{E}[\hat{\theta}_n] = \theta$.
- $\hat{\theta}_n$ is **consistent** (as an estimator of θ) if $\hat{\theta}_n \xrightarrow[n \rightarrow \infty]{} \theta$ (in some sense).
- $\bar{X}_n$ and $\hat{p}_n$ are unbiased and consistent estimators of μ and p .
- The **Standard Error (SE)** of an estimator is the standard deviation of its sampling distribution, $\widehat{SE}_{\bar{X}_n} = s_n/\sqrt{n}$, $\widehat{SE}_{\hat{p}_n} = \sqrt{\hat{p}_n(1 - \hat{p}_n)/n}$ (s_n is the sample standard deviation of $x_1, \dots, x_n$).
- The **Relative Standard Error (RSE)** of an estimator is the ratio of the Standard Error and the estimate, $\widehat{RSE}_{\bar{X}_n} = (s_n/\sqrt{n})/\bar{X}_n$, $\widehat{RSE}_{\hat{p}_n} = \sqrt{\hat{p}_n(1 - \hat{p}_n)/n}/\hat{p}_n$.
- The **Mean Squared Error (MSE)** of an estimator is the expected squared distance of the estimator to the parameter, $MSE_{\hat{\theta}_n} = \mathbb{E}[(\hat{\theta}_n - \theta)^2]$. If the estimator is unbiased, $MSE_{\hat{\theta}_n} = SE_{\hat{\theta}_n}^2$.

Confidence intervals

A **confidence interval** (CI) is a range of estimates for a parameter. Its *confidence level* $(1 - \alpha)$ represents the long-run proportion of CIs that contain the true value of the parameter.

- Approximate $(1 - \alpha)100\%$ CI on μ (the mean)

$$\left[\bar{x}_n - z_{\alpha/2} \frac{s_n}{\sqrt{n}}, \bar{x}_n + z_{\alpha/2} \frac{s_n}{\sqrt{n}} \right]$$

- Approximate $(1 - \alpha)100\%$ CI on p (a proportion)

$$\left[\hat{p}_n - z_{\alpha/2} \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}, \hat{p}_n + z_{\alpha/2} \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}} \right]$$

$$z_{\alpha/2} = \text{qnorm}(1 - \alpha/2)$$

Hypothesis testing

A **statistical test** consists in the confrontation of two statements about a probability distribution with the conclusion taken based on data.

- **Null hypothesis**, H_0 , represents the *default* situation. We will only reject it when the observations show strong evidence against it.
- **Alternative hypothesis**, H_1 . We will only regard it as correct (thus rejecting H_0) when the data show strong evidence favourable to it.

The **significance level** of a test, α , is the maximum probability of rejecting H_0 with random observations taken under it that we are about to tolerate.

The **p-value** is the *smallest level of significance at which H_0 can be rejected*. It is the probability that, under H_0 , a test result at least as extreme as the one actually observed is obtained.



S.2 Statistical analysis of simulated data (Ex. Area circle)

```
MC <- 1000 ; set.seed(1)
x <- runif(MC) ; y <- runif(MC) ; p <- sum(x^2+y^2 <= 1)/MC
prop.test(x = p*MC, n = MC, conf.level = 0.95)$conf.int[c(1,2)]
```

```
## [1] 0.7600308 0.8117295
```

```
t.test(x^2+y^2<=1,conf.level=0.95)$conf.int[c(1,2)] # not the best option
```

```
## [1] 0.7615804 0.8124196
```

```
(se <- sqrt(p*(1-p)/MC)) # standard error (depends on the units)
```

```
## [1] 0.01294724
```

```
p+qnorm(0.975)*se*c(-1,1)
```

```
## [1] 0.7616239 0.8123761
```

```
(rse <- se/p) # relative standard error (does not depend on the units)
```

```
## [1] 0.01645138
```

Example: Approximate the area of a circle of radius 1

```
4*p
```

```
## [1] 3.148
```

```
prop.test(x = p*MC, n = MC, conf.level = 0.95)$conf.int[c(1,2)]*4
```

```
## [1] 3.040123 3.246918
```

```
t.test(x^2+y^2<=1, conf.level=0.95)$conf.int[c(1,2)]*4 # not the best option
```

```
## [1] 3.046322 3.249678
```

```
4*(p+qnorm(0.975)*se*c(-1,1))
```

```
## [1] 3.046496 3.249504
```

```
print(c(4*se, rse)) # standard error and relative standard error
```

```
## [1] 0.05178896 0.01645138
```

S.3 Statistical validation techniques (GoF tests)

- **Kolmogorov-Smirnoff Goodness-of-Fit test** (continuous data, given F_0)

Test. $H_0 : F = F_0$ vs. $H_1 : F \neq F_0$

Test statistic. $D = \max_x |\hat{F}_n(x) - F_0(x)|$, with $\hat{F}_n$ empirical cdf.

The distribution of D (under H_0) does NOT depend on F_0 (continuous).

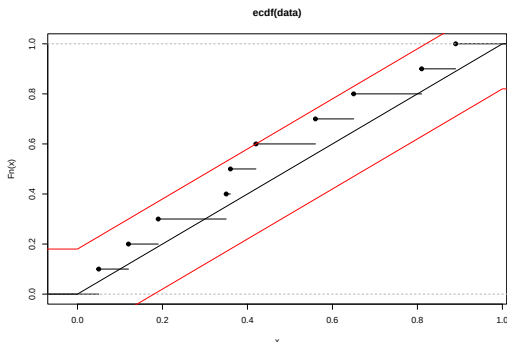
```
data <- c(.56, .35, .89, .81, .36, .19, .65, .42, .12, .05)
ks.test(x = data, y = "punif")$p.value
```

```
## [1] 0.8472806
```

The **p-value** of a statistical test is the smallest significance level that would lead to rejection of H_0 . It is the observed level of significance.

Statistical validation techniques (GoF tests)

```
plot(ecdf(data))  
aux <- seq(from = -.1, to = 1.1, by = 0.01)  
points(aux, punif(aux), type = "l")  
D <- ks.test(x = data, y = "punif")$statistic  
points(aux, punif(aux) + D, type = "l", col = "red")  
points(aux, punif(aux) - D, type = "l", col = "red")
```



Statistical validation techniques (GoF tests)

- **Chi-Square Goodness-of-Fit test** (discrete data, given F_0)

Test. $H_0 : F = F_0$ vs. $H_1 : F \neq F_0$

Test statistic. $X^2 = \sum_{i=1}^k \frac{(N_i - np_i)^2}{np_i}$,

k levels of the variable, N_i observed counts, and np_i expected counts.

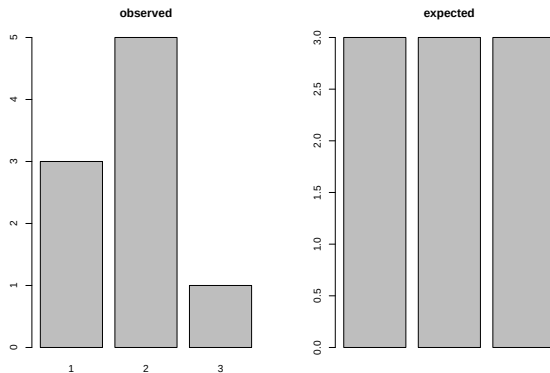
Under H_0 $X^2 \sim \chi_{k-1}^2$.

```
observed <- table(c(3, 2, 1, 2, 2, 1, 2, 1, 2))
probs <- rep(1/3, 3)
chisq.test(x = observed, p = probs)
```

```
##
## Chi-squared test for given probabilities
##
## data:  observed
## X-squared = 2.6667, df = 2, p-value = 0.2636
```

Statistical validation techniques (GoF tests)

```
n <- sum(observed)
par(mfrow = c(1,2))
barplot(observed, main = "observed")
barplot(probs*n, main = "expected")
```



Statistical validation (GoF tests, unspecified parameters)

- In order to test whether a sample was drawn from a *popular* family of distributions (e.g. normal), we can use specific tests (e.g. Shapiro-Wilk normality test, `shapiro.test`) or graphical tools (e.g. qq-plots, `qqnorm`).
- When some parameters are unspecified:
 - Continuous distribution, use KS test and simulate to estimate critical value or p-value.
 - Discrete distribution, use Chi-square test and subtract the number of estimated parameters from the degrees of freedom.
- The Chi-square test can be used with discrete data with infinitely many categories by grouping them into a finite number of categories and it is also often used with continuous data by grouping it into intervals.

Randomness tests

- **Wald-Wolfowitz runs test** of randomness for continuous data

Test. H_0 : random pattern of observations w.r.t. threshold (median)
vs. H_1 : pattern of observations is not random.

Test statistic. U : # runs (group of consecutive values either above or below threshold – median), N_1 (resp. N_2): # values above (below) threshold

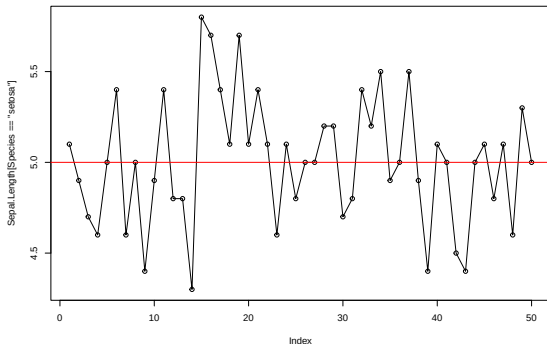
$$\frac{U - \left(\frac{2N_1N_2}{N_1+N_2} + 1 \right)}{\sqrt{\frac{2N_1N_2(2N_1N_2 - N_1 - N_2)}{(N_1+N_2)^2(N_1+N_2-1)}}}$$

```
# install.packages("randtests")  
require(randtests) ; data("iris")  
runs.test(iris$Sepal.Length[iris$Species=="setosa"])$p.value
```

```
## [1] 0.7428503
```

Randomness tests

```
attach(iris)
plot(Sepal.Length[Species=="setosa"], type = "l")
points(Sepal.Length[Species=="setosa"])
abline(h=median(Sepal.Length[Species=="setosa"]),col="red")
```



Independence tests

- **Ljung-Box test** of serial correlation.

Test. H_0 : independent (uncorrelated) observations vs.

H_1 : nonindependent observations (autocorrelations are not zero).

Test statistic. $Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$,

h number of lags tested, $\hat{\rho}_k$ sample autocorrelation at lag k .

Under H_0 $Q \sim \chi_h^2$.

```
Box.test(iris$Sepal.Length[iris$Species == "setosa"],  
         lag = 10, type = "Ljung")
```

```
##
```

```
## Box-Ljung test
```

```
##
```

```
## data: iris$Sepal.Length[iris$Species == "setosa"]
```

```
## X-squared = 9.5803, df = 10, p-value = 0.4781
```

Independence tests

```
acf(Sepal.Length[Species == "setosa"])
```

